Lecture 28

Tuesday, 25 April 2023 1:31 PM

Exam 2: Same format, open book, 1 pg front side,
Ch 5-7, conceptual 1-5, 8

→ 75 min, 100 pts

Electrolyte potential vs. Electrode potential

↓ This is what the R_Ω, R_{ct} analysis fit

↳ Resistive substrate (i.e. conductivity gradients)

(Electrolyte really close to electrode potential)

$t_+ \to 0$ assumption [fraction of current carried by species]

Binary electrolyte ← $t_+ \uparrow : 1$

$t_+ : 0$ ↓ Lots of other ions in species (conduction / diff.)

} Excess supporting electrolyte

} Shows up in flux in migration term ↳ dependent on conc. gradient)

↓ But need for tertiary current dist.

Electrochemical Reaction Engineering:

CSTR: $V_0(C_{A_0} - C_A) - r_A = 0$ (@ SS, isothermal)

↳ $\frac{-i_{ls}}{F} = 0$ ← surface area

$$\frac{-i_{ls}}{nF} = 0$$ → surface area

$$i = f\left(C_s, \frac{D}{\delta}\right)$$

$$\tau = \frac{V}{v_o} \quad ; \quad \sigma = \frac{s}{V}$$

σ: specific electrode area.

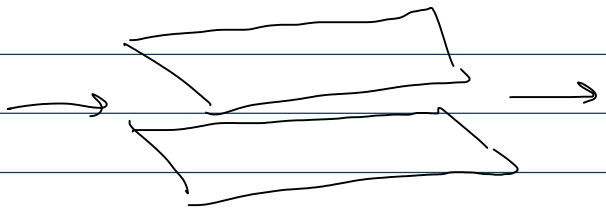
$$\frac{\sigma \tau i_A}{nF} = (C_{Ao} - C_A) \quad ; \quad \chi = \frac{C_{Ao} - C_A}{C_{Ao}}$$

K_c [=] $\frac{m}{s}$

$$\chi = \frac{\sigma \tau i_A}{C_A nF} \quad @ \text{ limiting current} \quad i_A = nF\left(\frac{D}{\delta}\right) C_A$$

$$\boxed{\chi = \frac{\sigma \tau K_L}{1 + \sigma i K_c}}$$

PFR:



$$\frac{i_A S}{nFL} = -v_o \frac{dC_A}{dx}$$